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SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO1 ASSESSMENT TOOL 2

2. Student Marks Analysis

A college maintains student marks in students.csv.

Task:

Read the dataset.

Calculate the average marks.

Identify students who scored below 40.

Add a new column named Result (Pass/Fail).

Export the updated file.

AIM: To read a CSV file containing student marks, calculate the average marks, identify students who scored below 40, add a Result (Pass/Fail) column, and export the updated dataset using Python Pandas.

PROCEDURE:

- Open Visual Studio Code.
- Create a project folder.
- Create the students.csv dataset.
- Install the Pandas library using `pip install pandas`.

- Create a Python file named marks_analysis.py.
- Import the Pandas library.
- Read the dataset using pd.read_csv().
- Calculate the average marks using mean().
- Display students who scored below 40.
- Add a new column named Result using Pass/Fail conditions.
- Export the updated dataset using to_csv().
- Execute the program and verify the output.

OUTPUT:

```

C:\Users\Vaandri\OneDrive\Desktop\StudentMarksAnalysis>python marks_analysis.py

Student Marks
  RollNo  Name  Marks
0    101  Alice    85
1    102   Bob    35
2    103  Charlie    72
3    104   David    28
4    105    Eva    91
5    106   Frank    55
6    107   Grace    39
7    108   Henry    64
8    109    Ivy    48
9    110   Jack    22

Average Marks = 53.9

Students Scored Below 40
  RollNo  Name  Marks
3    104   David    28
6    107   Grace    39
9    110   Jack    22

Updated Dataset
  RollNo  Name  Marks  Result
0    101  Alice    85    Pass
1    102   Bob    35    Fail
2    103  Charlie    72    Pass
3    104   David    28    Fail
4    105    Eva    91    Pass
5    106   Frank    55    Pass
6    107   Grace    39    Fail
7    108   Henry    64    Pass
8    109    Ivy    48    Pass
9    110   Jack    22    Fail
  
```

```

C:\Users\Vaandri\OneDrive\Desktop\StudentMarksAnalysis>python marks_analysis.py

Student Marks
  RollNo  Name  Marks
0    101  Alice    85
1    102   Bob    35
2    103  Charlie    72
3    104   David    28
4    105    Eva    91
5    106   Frank    55
6    107   Grace    39
7    108   Henry    64
8    109    Ivy    48
9    110   Jack    22

Average Marks = 53.9

Students Scored Below 40
  RollNo  Name  Marks
3    104   David    28
6    107   Grace    39
9    110   Jack    22

Updated Dataset
  RollNo  Name  Marks  Result
0    101  Alice    85    Pass
1    102   Bob    35    Fail
2    103  Charlie    72    Pass
3    104   David    28    Fail
4    105    Eva    91    Pass
5    106   Frank    55    Pass
6    107   Grace    39    Fail
7    108   Henry    64    Pass
8    109    Ivy    48    Pass
9    110   Jack    22    Fail

Updated file saved as students_updated.csv
  
```

RESULT: The student marks dataset was successfully analysed. The average marks were calculated, students scoring below 40 were identified, a Result column was added indicating Pass or Fail, and the updated dataset was exported to students_updated.csv.

6. Hospital Patient Records

A hospital maintains patient details in patients.csv.

Task:

Remove duplicate patient records.

Replace missing age values with the median age.

Identify patients older than 60.

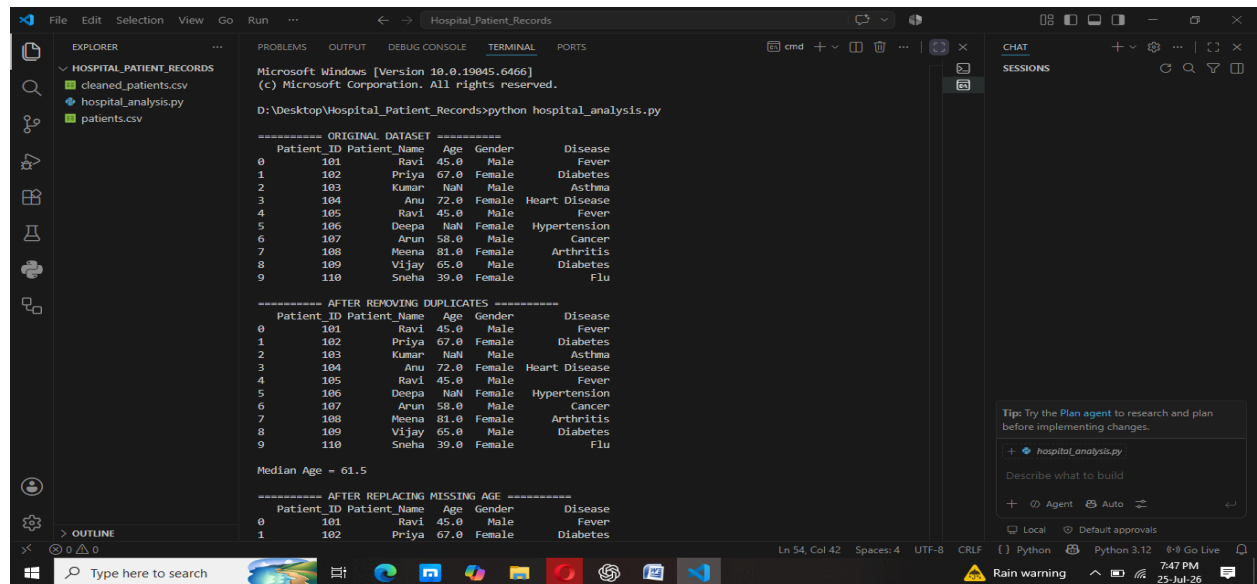
Save the cleaned dataset.

AIM: To clean the hospital patient dataset by removing duplicate records, replacing missing age values with the median age, identifying patients older than 60 years, and saving the cleaned dataset using Python and Pandas.

PROCEDURE:

- Open Visual Studio Code (VS Code) and create a project folder containing the `patients.csv` dataset and a Python file (`hospital_analysis.py`).
- Import the Pandas library and read the `patients.csv` file into a DataFrame.
- Remove duplicate patient records and replace missing values in the Age column with the median age.
- Filter the dataset to identify patients whose age is greater than 60 years.
- Save the cleaned dataset as `cleaned_patients.csv` using the `to_csv()` function.
- Execute the program in the VS Code terminal and verify the output and the generated cleaned dataset.

OUTPUT:



```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

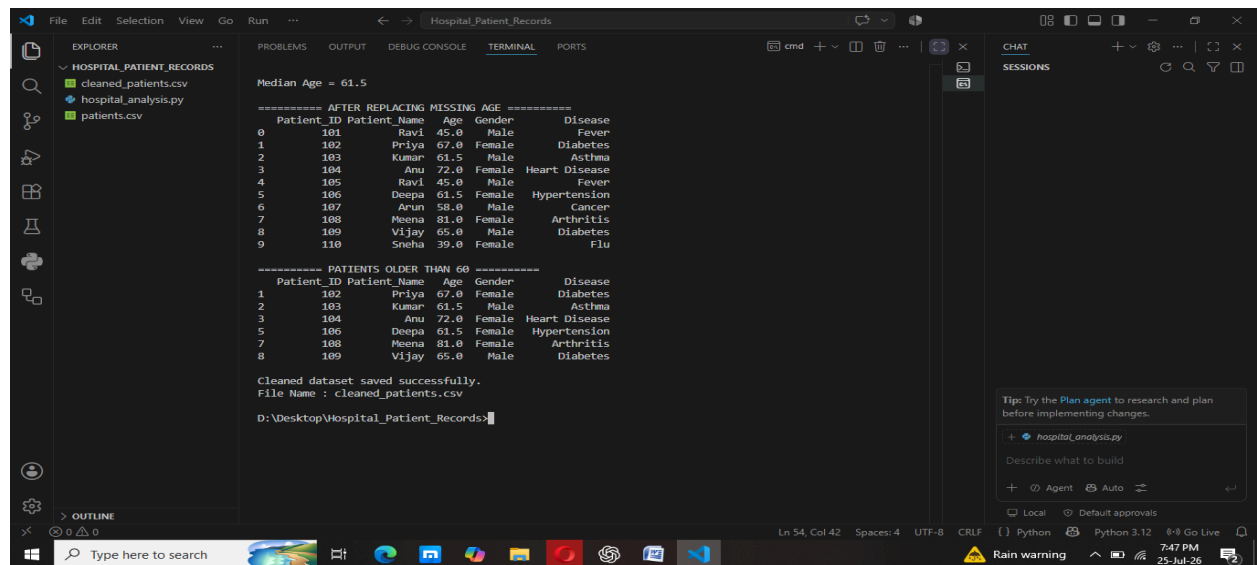
D:\Desktop\Hospital_Patient_Records>python hospital_analysis.py

===== ORIGINAL DATASET =====
Patient_ID Patient_Name Age Gender Disease
0 101 Ravi 45.0 Male Fever
1 102 Priya 67.0 Female Diabetes
2 103 Kumar NaN Male Asthma
3 104 Anu 72.0 Female Heart Disease
4 105 Ravi 45.0 Male Fever
5 106 Deepa NaN Female Hypertension
6 107 Arun 58.0 Male Cancer
7 108 Meena 81.0 Female Arthritis
8 109 Vijay 65.0 Male Diabetes
9 110 Sneha 39.0 Female Flu

===== AFTER REMOVING DUPLICATES =====
Patient_ID Patient_Name Age Gender Disease
0 101 Ravi 45.0 Male Fever
1 102 Priya 67.0 Female Diabetes
2 103 Kumar NaN Male Asthma
3 104 Anu 72.0 Female Heart Disease
4 105 Ravi 45.0 Male Fever
5 106 Deepa NaN Female Hypertension
6 107 Arun 58.0 Male Cancer
7 108 Meena 81.0 Female Arthritis
8 109 Vijay 65.0 Male Diabetes
9 110 Sneha 39.0 Female Flu

Median Age = 61.5

===== AFTER REPLACING MISSING AGE =====
Patient_ID Patient_Name Age Gender Disease
0 101 Ravi 45.0 Male Fever
1 102 Priya 67.0 Female Diabetes
```



```
Median Age = 61.5

===== AFTER REPLACING MISSING AGE =====
Patient_ID Patient_Name Age Gender Disease
0 101 Ravi 45.0 Male Fever
1 102 Priya 67.0 Female Diabetes
2 103 Kumar 61.5 Male Asthma
3 104 Anu 72.0 Female Heart Disease
4 105 Ravi 45.0 Male Fever
5 106 Deepa 61.5 Female Hypertension
6 107 Arun 58.0 Male Cancer
7 108 Meena 81.0 Female Arthritis
8 109 Vijay 65.0 Male Diabetes
9 110 Sneha 39.0 Female Flu

===== PATIENTS OLDER THAN 60 =====
Patient_ID Patient_Name Age Gender Disease
1 102 Priya 67.0 Female Diabetes
2 103 Kumar 61.5 Male Asthma
3 104 Anu 72.0 Female Heart Disease
5 106 Deepa 61.5 Female Hypertension
7 108 Meena 81.0 Female Arthritis
8 109 Vijay 65.0 Male Diabetes

Cleaned dataset saved successfully.
File Name : cleaned_patients.csv

D:\Desktop\Hospital_Patient_Records>
```

RESULT: The hospital patient dataset was successfully processed using Python and the Pandas library. Duplicate patient records were removed, missing age values were replaced with the median age, patients older than 60 years were identified, and the cleaned dataset was successfully saved as `cleaned_patients.csv`. The data cleaning process was completed successfully, ensuring a consistent and accurate dataset for further analysis.